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$$1. \text{ 令 } z = x + yi$$

$$\therefore f(z) = \begin{cases} \frac{xy}{x^2+y^2}, & z \neq 0 \\ 1, & z = 0. \end{cases}$$

当直线沿 $y=kx$ 趋于 0 时,

$$f(z) = \frac{x \cdot kx}{x^2 + kx^2} = \frac{k}{1+k} \text{ 随 } k \text{ 值变化}$$

$\therefore \lim_{z \rightarrow 0} f(z)$ 不存在,

故 $f(z)$ 在原点不连续

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2、

$$w = \frac{x - yi}{x^2 + y^2}$$

$$= \frac{(x - yi)(x + yi)}{(x^2 + y^2)(x + yi)}$$

$$= \frac{1}{x + yi}$$

$$= \frac{1}{z}$$

$$\therefore \frac{1}{z} w = u + iv$$

$$\therefore z = \frac{1}{u + iv} = \frac{u - iv}{u^2 + v^2}$$

$$\therefore \frac{u}{u^2 + v^2} = x, \quad y = \frac{-v}{u^2 + v^2}$$

$$\therefore \left(\frac{u}{u^2 + v^2}\right)^2 + \left(\frac{-v}{u^2 + v^2}\right)^2 = 1$$

$$\therefore (u - u^2 - v^2)^2 + v^2 = (u^2 + v^2)^2$$

$$\therefore u^2 + v^2 - 2u(u^2 + v^2) = 0$$

$$(1 - 2u)(u^2 + v^2) = 0$$

$$\therefore u = \frac{1}{2}, \text{ 直线 }.$$

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3. $z = x + yi$

$$f(z) = e^{|z|^2} = e^{x^2+y^2}$$

$$= u(x, y) + i v(x, y)$$

$$u(x, y) = e^{x^2+y^2}, \quad v(x, y) = 0.$$

$$\therefore u'_x = 2x e^{x^2+y^2}, \quad v'_x = 0$$

$$u'_y = 2y e^{x^2+y^2}, \quad v'_y = 0$$

由CR条件 $\begin{cases} u'_x = v'_y \\ u'_y = v'_x \end{cases}$

$$\therefore \begin{cases} x=0 \\ y=0 \end{cases}$$

$\therefore f(z)$ 在 $z=0$ 处可导, 处处不可导

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$$\begin{aligned} 4. \quad & \text{令 } z = e^{i\theta} \\ & \therefore |dz| = |de^{i\theta}| = |ie^{i\theta} d\theta| \\ & = d\theta \\ & = \frac{dz}{iz} \end{aligned}$$

$$\begin{aligned} & \therefore \int_{|z|=1} \frac{1}{|z-2|^2} |dz| \\ & = \int_{|z|=1} \frac{1}{(z-2)(\bar{z}-2)} \times \frac{1}{iz} dz \end{aligned}$$

$$= -i \int_{|z|=1} \frac{1}{z(5-2z-\frac{2}{z})} dz$$

$$= -i \int_{|z|=1} \frac{1}{-2z^2 + 5z - 2} dz.$$

$$= i \int_{|z|=1} \frac{1}{(z-2)(2z-1)} dz.$$

在 $|z|=1$ 内仅有 $z=\frac{1}{2}$ 这个点

$$\begin{aligned} \therefore i \int_{|z|=1} \frac{1}{(z-2)(2z-1)} dz &= i 2\pi i \operatorname{Res}[f(z), z=\frac{1}{2}] \\ &= -2\pi \times \frac{1}{\frac{1}{2}-2} \\ &= -2\pi \times -\frac{2}{3} = \frac{4}{3}\pi \end{aligned}$$

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5. 解

同 19.A

6.

$\sum_{n=0}^{\infty} a_n z^n$ 收敛半径为 R .

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

即证: $|a_n| \cdot r^n \leq M(r)$

$$|a_n| = \left| \frac{f^{(n)}(z_0)}{n!} \right| = \left| \frac{1}{2\pi i} \int \frac{f(z)}{(z-z_0)^{n+1}} dz \right|$$

$$\leq \left| \frac{1}{2\pi i} \right| \max_{0 \leq \theta \leq 2\pi} |f(re^{i\theta})| \int_{|z|=r} \frac{|dz|}{|z|^{n+1}}$$

$$z = r \cdot e^{i\theta},$$

$$dz = ir e^{i\theta} d\theta$$

$$\frac{|dz|}{|z|} = \frac{|z| d\theta}{|z|}$$

$$= \left| \frac{M(r)}{2\pi i} \right| \int_0^{2\pi} \frac{r i r e^{i\theta} d\theta}{r \cdot e^{i\theta}} \cdot \frac{1}{r^{n+1}}$$

$$= \frac{M(r)}{2\pi r^n} \int_0^{2\pi} d\theta$$

$$= \frac{M(r)}{2\pi r^n} \times 2\pi$$

$$= \frac{M(r)}{r^n}$$

得证.

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7. 解

同 19 A

$$f(z) = -i(z-1)^2$$

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8. 解

$$\text{令 } f(z) = \frac{a}{z+1} - \frac{b}{z+2}$$

$$\therefore f(z) = \frac{az + 2a - bz - b}{(z+1)(z+2)}$$

$$\therefore \begin{cases} a - b = 1 \\ 2a - b = 0 \end{cases}$$

$$\therefore \begin{cases} a = -1 \\ b = -2 \end{cases}$$

$$\therefore f(z) = \frac{2}{z+2} - \frac{1}{z+1}$$

$$= 2 \times \frac{1}{(z-2)+4} - \frac{1}{(z-2)+3}$$

$$= \frac{2}{4} \times \frac{1}{1 - (-\frac{z-2}{4})} - \frac{1}{3} \left(\frac{1}{1 - (-\frac{z-2}{3})} \right)$$

$$= -\frac{1}{3} \sum_{n=0}^{\infty} \left(-\frac{z-2}{3}\right)^n + \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{z-2}{4}\right)^n$$

$$= \sum_{n=0}^{\infty} \left[(-1)^{n+1} \cdot 3^{-n-1} + \frac{1}{2} (-1)^n 4^{-n} \right] (z-2)^n$$

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$$9. \quad f(z) = \frac{1}{(z-3)(z-2)} = \frac{1}{z-3} - \frac{1}{z-2}$$

∴ (1). $0 < |z-2| < 1$ 时,

$$\begin{aligned} f(z) &= -\frac{1}{z-2} + \frac{1}{(z-2)-1} \\ &= -(z-2)^{-1} - \frac{1}{1-(z-2)} \\ &= -(z-2)^{-1} - \sum_{n=0}^{\infty} (z-2)^n \\ &= -\sum_{n=-1}^{\infty} (z-2)^n \end{aligned}$$

$$\begin{aligned} (2). \quad f(z) &= \frac{1}{z-3} - \frac{1}{(z-3)+1} \\ &= (z-3)^{-1} - \frac{1}{z-3} \left(\frac{1}{1 + \frac{1}{z-3}} \right) \\ &= (z-3)^{-1} - (z-3)^{-1} \sum_{n=0}^{\infty} (-1)^n (z-3)^{-n} \end{aligned}$$

$$\int_{|z-2|=1/2} f(z) dz = 2\pi i C_{-1} = 2\pi i \times -1 = -2\pi i$$

$$\begin{aligned} \int_{|z-3|=2} f(z) dz &= 2\pi i C_{-1} = 2\pi i (1 - 1 \times 1) \\ &= 0 \end{aligned}$$

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10. 解 该函数极点 $z_k^6 = -1$, $z=0$ 在 $|z|=2$ 内,

$z=4$ 在 $|z|=2$ 外.

$$\int_{|z|=2} \frac{z^5 \cos \frac{1}{z}}{(1+z^6)(z-4)} = 2\pi i \left[\operatorname{Res} \left[\frac{\frac{1}{z^5} \cos z}{(1+\frac{1}{z^6})(\frac{1}{z}-4)} \right], 0 \right]$$

$$- \operatorname{Res} \left[\frac{z^5 \cos \frac{1}{z}}{(1+z^6)(z-4)}, 4 \right]$$

$$= 2\pi i \times \left[\operatorname{Res} \left(\frac{\cos z}{(1+z^6)(1-4z)}, 0 \right) - \frac{4^5 \cos \frac{1}{4}}{(1+4^6)} \right]$$

$$= 2\pi i \times - \frac{4^5 \cos \frac{1}{4}}{1+4^6}$$

$$= - \frac{2 \times 4^5}{1+4^6} \pi i \cos \frac{1}{4}$$

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11. 解

$$\int_0^{+\infty} \frac{x \sin mx}{(x^2+a^2)^2} dx = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{x \sin mx}{(x^2+a^2)^2} dx$$

$$= \frac{1}{2} \operatorname{Im} \int_{-\infty}^{+\infty} \frac{z \cdot e^{imz}}{(z^2+a^2)^2} dz.$$

对 $f(z) = \frac{z \cdot e^{imz}}{(z^2+a^2)^2}$, 极点为 $z = \pm ai$,

在 x 轴上方仍有 $z = ai$

$\operatorname{Res}[f(z), z=ai]$

$$= \frac{1}{(2-1)!} \times \left[(z-ai)^2 \frac{ze^{imz}}{(z-ai)^2(z+ai)^2} \right]' \Big|_{z=ai}$$

$$= \left(\frac{ze^{imz}}{(z+ai)^2} \right)' \Big|_{z=ai}$$

$$= \frac{(e^{imz} + imze^{imz})(z+ai)^2 - 2(z+ai)(ze^{imz})}{(z+ai)^4} \Big|_{z=ai}$$

$$= \frac{e^{imz}(1+imz)(z+ai) - 2ze^{imz}}{(z+ai)^3} \Big|_{z=ai}$$

$$= \frac{e^{-am}(1-im)a - 2ae^{-am}}{8a^3i^3}$$

$$= \frac{-2ai \times ma \times e^{-am} + 84a^2i}{8a^3i^3}$$

$$= \frac{me^{-am}}{4a}$$

$$\int_{-\infty}^{+\infty} f(z) dz = 2\pi i \times \frac{me^{-am}}{4a}$$

$$\therefore \text{原式} = \frac{1}{2} \times 2\pi i \times \frac{me^{-am}}{4a}$$

$$= \frac{\pi m}{4a e^{am}}$$

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